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RESEARCH ARTICLE

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Overcoming barriers: the potential of AI digital textbooks in multicultural middle school education

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ABSTRACT

This study explores the potential of AI Digital Textbooks (AIDT) to improve educational experiences for multicultural middle school students in Korea. As the student population becomes more diverse, traditional methods often fail to meet the unique needs of multicultural learners. The research evaluates AIDT's effectiveness in providing personalized, interactive, and culturally relevant content. By analyzing test versions of AIDT through inclusive education theory, the study identifies gaps, including the lack of multilingual support, culturally relevant content, and adaptive learning pathways. The findings stress the need for real-time feedback, interactive simulations, personalized study plans, collaborative learning tools, and enhanced cultural understanding through group projects. The study also emphasizes the importance of comprehensive teacher training to integrate AIDT effectively in diverse classrooms. The recommendations aim to create a more inclusive and supportive environment, ensuring that both Korean and multicultural students can thrive academically and socially.

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AI digital textbooks; multicultural education; academic achievement; adaptive learning tools; middle school student

Artificial Intelligence (AI) technology has advanced rapidly across various sectors, including education. Its integration into learning environments has changed how students access educational content and how teachers facilitate learning and classroom management (Hwang et al., 2020; Idroes et al., 2023; Yu et al., 2024). In the Republic of Korea (Korea), the development of AI Digital Textbook (AIDT), planned to begin in 2025, represents a major step towards digital innovation in education.

Despite a decline in Korea's overall student population, the proportion of multicultural students has steadily increased, comprising 3.8% of elementary and secondary school enrolment in 2023. This includes 4.4% in elementary schools, 3.3% in middle schools, and 1.7% in high schools (MoE & KEDI, 2024). In 350 schools nationwide, over 30% of students come from immigrant backgrounds (Yoon et al., 2023). High-density multicultural schools, where multicultural students make up 70 to 80% of the population, are concentrated in regions like Gyeonggi and Chungcheong (Kim, 2024). These schools face unique challenges, as students often struggle to adapt, and traditional teaching methods prove insufficient (Yoon et al., 2023).

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As Korea's student demographics become more diverse, AI-driven educational tools must address the unique needs of multicultural students, especially those encountering linguistic and cultural barriers. AIDT features such as personalized learning pathways, interactive feedback, and adaptive content are crucial for providing language support and culturally relevant materials (MoE, 2023). Despite the Korean government's under the motto '모두를 위한 맞춤 교육' [Personalized Education for All], gaps remain in meeting the challenges faced by multicultural students.

This research explores how AI textbooks can promote educational equity by fostering a more inclusive and supportive learning environment. The study examines publicly accessible AIDT test versions for middle school students using evaluative analysis and inclusive education theory to pinpoint current characteristics, barriers, and difficulties encountered by multicultural pupils as well as methods to improve inclusion.

Inclusive education theory

Inclusive education aims to include all students, regardless of their physical, intellectual, social, emotional, linguistic, or other conditions (UNICEF, 2015). Historically, it emerged as an effort to integrate students with disabilities, focusing on removing barriers to access and participation (Graham, 2020). Since the 1970s, countries like the United States, the United Kingdom, Canada, and Australia have expanded this concept to address the educational rights of marginalized groups, including African American students (Rapp & Corral-Granados, 2024). This evolution underscores the global struggle for educational equity and inclusion.

Contemporary efforts to advance inclusive education, influenced by historical foundations, have been shaped by international organizations such as the United Nations International Children's Emergency Fund (UNICEF). UNICEF emphasizes the fundamental right of every child to access quality education and highlights the significant challenges faced by children with disabilities and other marginalized groups in mainstream education (UNICEF, n.d.). UNICEF's multifaceted approach to promoting inclusive education involves capacity-building, advocacy, awareness-raising, and implementation assistance. It emphasizes the importance of structural reforms across various levels, such as policy alignment, teacher training, and community engagement, to effectively integrate all children into mainstream educational environments (UNICEF, n.d.).

The UN's Sustainable Development Goal (SDG) of 'Leave No One Behind (LNOB)' emphasizes eliminating educational disparities, particularly for marginalized groups, including multicultural students (UNSDG, n.d.). LNOB, a core commitment of the 2030 Agenda for Sustainable Development, aims to eradicate poverty, discrimination, and inequality by prioritizing the marginalized. In education, LNOB highlights the importance of creating inclusive systems that support all learners (UNSDG, n.d.). Inclusive education theory stresses that all students, regardless of their background, should have access to equitable learning opportunities (UNICEF, 2015). AIDT holds promise in this regard by incorporating adaptive learning features that could cater to diverse learning needs. For instance, personalized learning pathways in AIDT could ensure that multicultural students, who may struggle with the Korean language, receive tailored educational content that meets their proficiency

levels. This approach aligns with the principle of *accessibility* by ensuring that all students can engage with the material, irrespective of their linguistic background.

Multicultural middle schoolers' adversities

The number of multicultural students in Korea has steadily increased. According to the Korea Education Development Institute (KEDI, 2021), 98.1% of multicultural children enrol in elementary school, but this drops to 87.9% by high school, contrasting with the national average of 92.4%. Moreover, only 49.5% of multicultural students pursue higher education, compared to 67.6% of Korean students (MoE & KEDI, 2024).

One primary reason for this decline is the difficulty multicultural students face in adapting to the school environment and curriculum. Many struggles with Korean, which is essential for academic success. Studies show that 64% of multicultural students cite difficulty with schoolwork, while 54% report challenges in socializing with peers (KESS, 2018; Park & Chung, 2021). In summary, their struggles with the Korean language hinder both academic performance and social integration, isolating them and preventing progression to high school.

Research shows that language proficiency is crucial for academic success; students who struggle with the primary language of instruction often fall behind (Christensen & Stanat, 2007). For multicultural students in Korea, insufficient support in Korean language acquisition fosters a cycle of disengagement and lower performance, leading to some leaving school before high school (Kim, 2020). Furthermore, these students often face social exclusion, which exacerbates isolation and academic challenges (Vo et al., 2018).

Research on multicultural education in countries like Canada shows that targeted language support programmes are crucial for helping immigrant students succeed academically and socially by bridging language gaps and offering culturally relevant learning experiences (Vo et al., 2018). Implementing similar programmes in Korea could improve retention and progression rates for multicultural students by providing the linguistic and socio-emotional support needed to thrive in middle and high school. These challenges are particularly evident in middle school, where strong societal pressure to graduate and pursue higher education can leave many multicultural students either not advancing to high school or entering labour-intensive occupations.

This process is partly driven by Korea's competitive education system, which places immense pressure on academic achievement as a pathway to stable employment and social mobility. These high expectations create significant challenges for multicultural students, often leading to early disengagement from both school and society. Research shows these students are more likely to enter lower-skilled, lower-paying jobs due to insufficient academic qualifications (Choi, 2021; Hong, 2010). As a result, many end up in labour-intensive industries with limited opportunities for advancement, reinforcing a cycle of socioeconomic disadvantages.

This issue is worsened by the perception that multicultural students, especially those from immigrant or non-Korean backgrounds, lack the academic drive or suitability for traditional education (Asia Society, n.d.). The bias often limits available support to help these students overcome their challenges. In contrast, countries like Canada offer targeted support, such as after-school tutoring and language classes, which improve high school retention and post-secondary aspirations among immigrant students (Christensen

& Stanat, 2007; Vo et al., 2018). Adopting similar measures in Korea could help multicultural students continue their education beyond high school and avoid low-wage labour.

Without comprehensive language and academic support, multicultural students may struggle with social integration, crucial for building a support network that fosters educational and professional aspirations. The article *How Korean Language Students Studying in Seoul Navigate Linguistic and Cultural Homogeneity* (Brown, 2024) highlights how non-Korean students face racialized experiences, limited interaction with native speakers, and challenges in language proficiency. Even students of Korean heritage or appearance face difficult expectations regarding language skills and cultural norms. This underscores the need for a more inclusive, multicultural approach in study-abroad programmes in Korea. Research shows that fostering a sense of belonging boosts motivation for higher education (Christensen & Stanat, 2007). Implementing culturally responsive teaching, mentorship, and community involvement could help multicultural students envision and pursue career paths beyond labour-intensive jobs.

A national survey by the Ministry of Gender Equality and Family (2022) found that while 62% of multicultural students aspired to attend a four-year university, only 40.5% enrolled in higher education. The low enrolment rate is mainly due to language barriers, which limit educational information for multicultural parents (Ministry of Gender Equality and Family, 2022). The survey also revealed that 50.4% of immigrant and naturalized parents cited *guidance and management of studies* as their biggest challenge, followed by *lack of information on further education and career paths* at 37.6%. This underscores the importance of AIDT as a vital support for linguistically and academically disadvantaged students.

In summary, while efforts are being made to improve educational opportunities for multicultural students in Korea, significant challenges remain. To address this challenge, the Korean Ministry of Education could partner with AI developers to integrate language support tools in the AIDT system, providing real-time translation and context-based language scaffolds. These tools would help multicultural students overcome language barriers, aligning with the Korean government's 'Personalized Education for All' initiative, and offering a model for multilingual education.

Korean artificial intelligence digital textbooks

Korea is the first country to announce the use of AIDT. As societies transition into the digital age, the Korean Ministry of Education has actively promoted enhanced textbooks as part of broader educational reforms (MoE, 2024). This initiative aims to improve individual student learning outcomes and foster inclusivity. According to the MoE (2024), AIDT will be implemented in 2025, focusing on priority subjects such as Mathematics, English, Informatics, and Korean (Special Education). The rollout will target 3rd and 4th graders in elementary school, as well as freshmen in middle school and high school. This pioneering effort presents Korea with a unique opportunity to become a global leader in the development and implementation of AIDT, setting a benchmark for other nations and demonstrating its commitment to educational innovation on the world stage.

In the past, digital supplemental materials for bilingual learners have been available, but only for elementary school students and subjects like social studies and Korean language. Although called digital textbooks, these resources still require teachers to print PDF files daily. In contrast, AIDT uses artificial intelligence technology to provide a more integrated educational experience, supporting all key participants-students, teachers, and parents-more effectively than traditional paper textbooks.

According to the Korea Educational Research Information Service (KERIS, n.d.), the core services of AIDT encompass a range of educational support tailored to different stakeholders, as outlined below:

- **For students**, AIDT provides comprehensive learning diagnostics, adaptive paths, personalized content recommendations, and individualized support through an AI tutor.
- **For teachers**, AIDT offers lesson design tools and AI assistant support, including content restructuring, tailored resources, and data-driven management to track student progress.

As a pioneer in the field, Korea has the unique opportunity to harness artificial intelligence to create customized learning experiences that adapt to each student's needs. For multicultural learners, AI can provide personalized language support, tailored content, and culturally relevant examples, enhancing both comprehension and engagement (Son et al., 2023). This adaptability enables AI to bridge gaps for students with diverse language proficiencies, cultural backgrounds, and academic skills, making education more accessible and inclusive (Choi, 2024; Son et al., 2023; Wang et al., 2023). By aligning educational technology with the needs of diverse students, Korea has the chance to lead the way in closing achievement gaps and fostering equitable learning environments, setting a global example for others to follow.

Methods

This study conducted a detailed analysis of a test version of middle school AIDT and teacher support through the lens of inclusive education theory, focusing on how well it addresses the unique needs of multicultural middle school students.

Data collection

In this study, we evaluated publicly available AIDT test versions from various subjects, selecting English, Social Studies, and Science for middle school students from the webpage at https://dtbook.edunet.net/. These subjects were chosen to represent a cross-section of curricular content, where multicultural middle school students might experience academic language and cultural barriers. We selected one representative chapter from each subject, English, social studies, and science, focusing on student textbooks that feature a range of features and content types, including text, images, and interactive elements.

We applied three key criteria for analysis: content accessibility, diversity through personalization, and teacher support mechanisms. These criteria were developed to assess how well the AIDT addresses the unique challenges faced by multicultural students, such as language barriers, social integration, and differentiated learning

needs. Images of some parts of the selected chapters were captured using screen capture tools, ensuring clarity and legibility. This comprehensive approach facilitated detailed analysis of the AIDT content, focusing on its effectiveness in helping students overcome language and cultural barriers in academic settings. It is important to note that the AIDT webpage is still in the testing phase, so some details may change before full implementation.

For the teachers' support, we were only able to access *The Guide to Using Digital Textbooks for Middle School Teachers* published by the MoE and KERIS (2018), since the researchers lacked the registered teacher licence numbers required to access additional materials for teachers on the website. However, this reliance on a single source indicates that the study's evaluation of teacher support mechanisms within the AIDT framework is based on one authoritative guide.

Analysis criteria

The analysis evaluated the effectiveness of AIDT in supporting students in overcoming language and cultural barriers within academic settings, aligning its framework with international best practices in inclusive education. The analysis criteria were based on the three core principles of inclusive education theory: accessibility of content, diversity through personalization, and support, all under the umbrella of equity from the Korean MoE and the United Nations (UN) as shown in Figure 1.

Content accessibility

This criterion focused on the extent to which the AIDT provides multilingual support and culturally relevant content. The evaluation involved a comparative analysis of the AIDT's features against those implemented in countries known for strong language and cultural

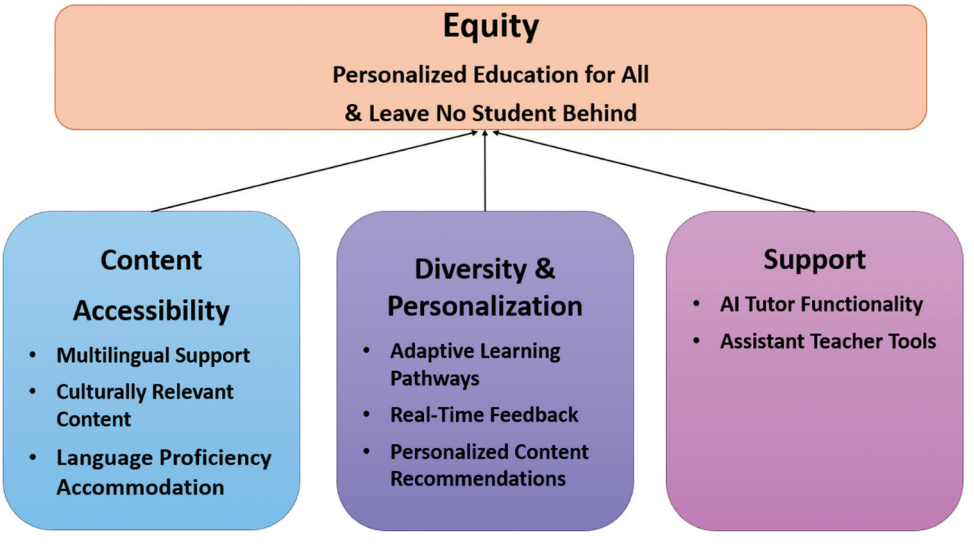


Figure 1. Criteria for evaluating AIDT based on inclusive education theory.

programmes, such as Canada (Evans, 2019; Janakiraman et al., 2019). Specifically, the analysis assessed:

- **Multilingual Support**: This criterion assessed the availability of content in multiple languages, ease of switching between them, and the presence of translation tools, bilingual resources, and language-specific modules.
- **Culturally Relevant Content**: This criterion focused on materials that reflect the cultural backgrounds of diverse students, including literature, historical perspectives, and cultural practices that acknowledge and celebrate diversity.
- **Language Proficiency Accommodation**: This criterion assessed how AIDT supports students with varying language proficiency, focusing on scaffolding techniques such as simplified language options, visual aids, and interactive elements to enhance comprehension and engagement.

Diversity through personalization

This criterion analysed AIDT's ability to personalize learning experiences to meet the diverse needs of multicultural students (Lindner et al., 2019; Shemshack & Spector, 2020). Key aspects evaluated included:

- **Adaptive Learning Pathways**: This criterion assessed the system's ability to adjust the content difficulty and based on individual student performance and learning pace, including the use of algorithms to tailor educational experiences.
- **Immediate and Actionable Feedback**: This criterion focused on providing immediate, actionable feedback to help students understand their progress and identify areas for improvement. It was assessed by examining the types of feedback (e.g. textual, visual, auditory) and their relevance to students' cultural and academic contexts.
- **Personalized Content Recommendations**: This criterion assessed the system's ability to suggest learning materials and activities that align with the student's cultural background and academic needs, focusing on the diversity of recommendations and their alignment with the student's learning history and preferences.

Support mechanisms

This criterion examined the tools and functionalities available within the AIDT to assist teachers in facilitating student learning (Rapp & Corral-Granados, 2024; Shemshack & Spector, 2020). The evaluation focused on:

- **AI Tutor Functionality**: This criterion assessed the effectiveness of AI-driven tutoring features in providing personalized academic support, focusing on the AI tutor's ability to offer explanations, answer questions, and provide practice exercises tailored to the student's learning needs.

- **Teacher Support Tools**: This criterion assessed tools designed to assist teachers in managing multicultural classrooms, focusing on features such as lesson planning aids, student progress tracking, and resources for addressing cultural and linguistic diversity.

Limitations

The limitations of this research stem from the test version of the AIDT, which may not fully reflect the final product's capabilities, potentially affecting the evaluation accuracy. Additionally, due to the large number of textbook publishers, it was not feasible to review all of them, limiting the generalizability of our findings across all digital AI textbooks.

Despite these limitations, the research offers significant benefits. The developed criteria offer a structured approach to assessing AIDT's effectiveness in supporting multicultural students, serving as a valuable tool for future evaluations. The insights gained lay the groundwork for future studies and can help identify potential errors in government guidelines, improving the effectiveness of AIDT on multicultural education.

Results and discussion

The publicly accessible test versions of AIDT for middle school students include subjects like English, social studies, and science, with several versions from different publishers. To ensure diverse representation, we selected English for 7th graders, Social Studies for 8th graders, and Science for 9th graders, all from the same publisher, Mirae N. This approach enabled a comprehensive analysis of how AIDT supports multicultural middle school students across various academic contexts. Additionally, our study complied with the publisher's criteria for using their content without prior permission, ensuring ethical and legal use of the AI Digital Textbooks (Mirae N, 2021).

English for 7th graders

The English AIDT featured multimedia resources such as audio files, video files, speaking and writing activities, and bilingual (Korean-English) interpretations. Notably, speaking and writing activities provided interchangeable options, enabling students to choose their answers based on their preferences. Supplementary materials for fast finishers were also included along with self-checking sections to allow students to monitor their progress.

Despite these interactive features, the English AIDT exhibited significant shortcomings in supporting multicultural students. The analysis showed the English textbook lacked translation tools and bilingual resources hindered non-native Korean speakers' ability to understand the material. Furthermore, the absence of culturally diverse examples reduced its relevance and engagement for multicultural learners. Minimal scaffolding, such as simplified language and visual aids, posed challenges for students with limited proficiency, as shown in Figure 2.

Figure 2 highlights the types of interactive exercises designed to enhance students' language skills. Despite its interactive features, the English AIDT demonstrated significant gaps in inclusivity and adaptability for multicultural learners. In terms of personalization, the AIDT did not dynamically adjust content difficulty based on student performance. This limitation may restrict its ability to accommodate students at various levels of proficiency. The English AIDT did not adjust content difficulty based on student performance, making it less effective for accommodating varying learning paces and styles.

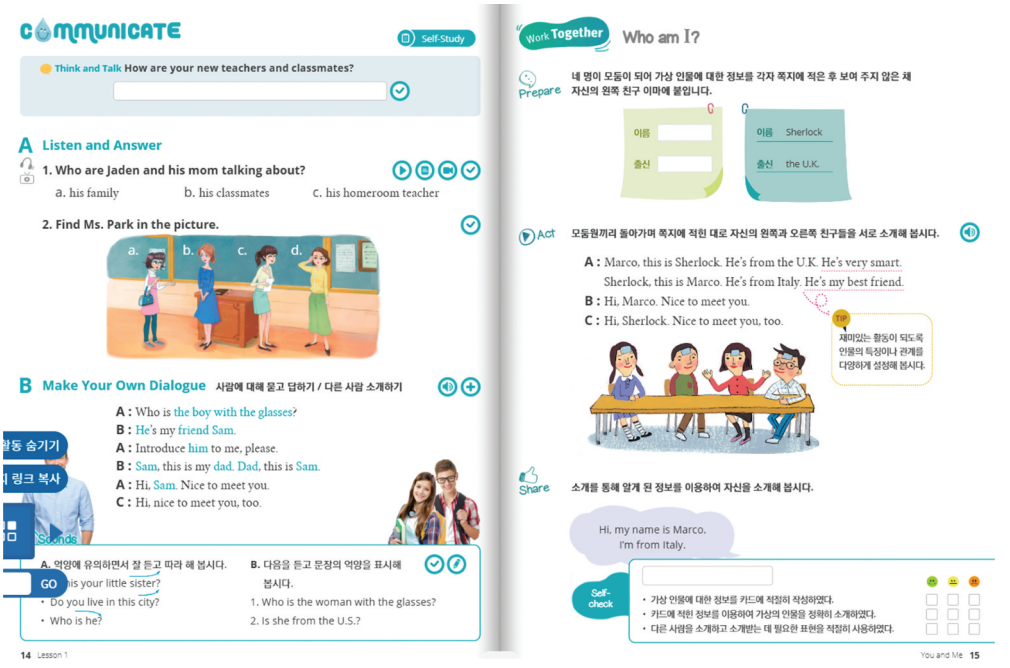


Figure 2. Communication activities from the English AIDT (Choi et al., 2018, pp. 14–15).

Moreover, feedback within the textbook was limited, textual, and lacked cultural customization essential for guiding multicultural students in improving their language skills. Effective feedback should be immediate, actionable, and culturally relevant to help students understand their progress and areas for improvement. The lack of such feedback in the English AIDT means that students may not receive the guidance they need to enhance their learning outcomes as shown in Figure 3.

Figure 3 demonstrates the inclusion of translated texts aimed at aiding comprehension. Despite this feature, the analysis found that the English AIDT did not dynamically adjust content difficulty based on student performance, limiting its effectiveness for students with varying proficiency levels.

Lastly, there was no evidence of content recommendations aligned with students' cultural backgrounds or language levels, limiting engagement for multicultural students. Personalized content recommendations can help make learning more relevant and engaging by connecting the material to students' own experiences and interests. The absence of this feature in the AIDT suggests that it may not fully support the diverse needs of its users. Overall, the findings indicate that while the English AIDT has potential, it currently falls short in several key areas needed to effectively support multicultural students.

Social studies for 8th graders

The Social Studies AIDT focused on Korean society, covering topics such as the Korean Constitution, labour law, and related subjects. As social studies often involve complex

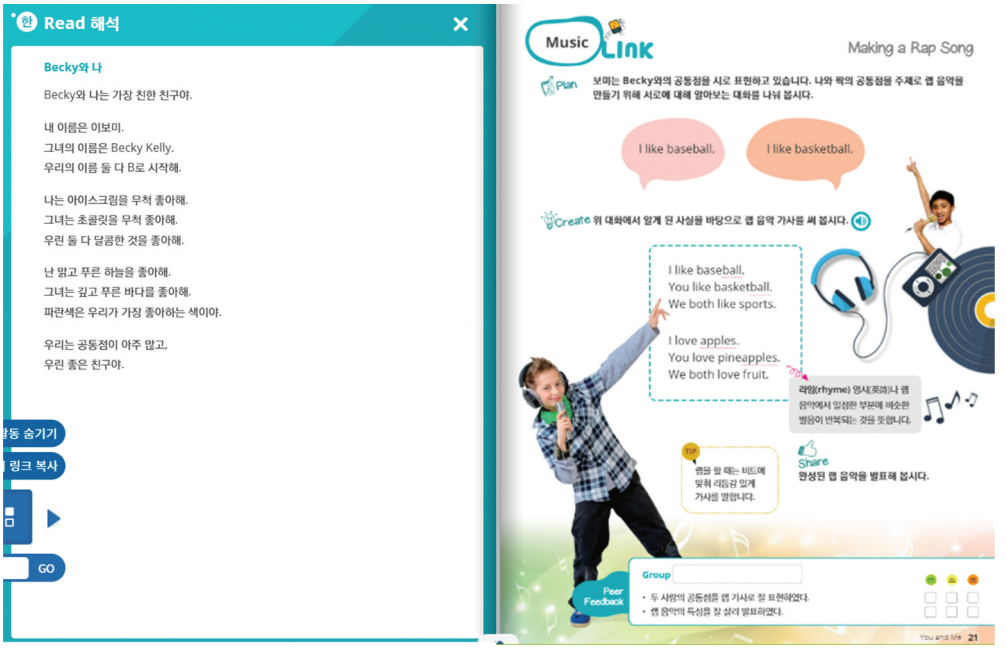


Figure 3. Translated reading materials from the English AIDT (Choi et al., 2018, pp. 20–21).

academic terms and Sino-Korean vocabulary, substantial background knowledge is required for comprehension (Kim, 2020; Kim & Cho, 2024). All materials were written and explained entirely in Korean, without additional multilingual or supplementary resources, as shown in Figure 4.

The analysis revealed that the absence of translation tools or bilingual options creates a significant barrier for non-native Korean speakers, particularly for understanding advanced social studies concepts in the Social Studies AIDT. Despite the potential to explore global comparisons or contrasts, the content failed to include diverse cultural perspectives, limiting its inclusiveness and engagement. Additionally, the absence of simplified language or interactive scaffolds made it challenging for students with lower Korean proficiency to engage with the content, as shown in Figure 4.

Figure 4 shows how the material is structured to provide essential information and summaries in the chapter related to Korean labour law. However, the analysis revealed that the textbook lacked multilingual support and culturally relevant content, which were critical for making the material accessible and engaging for multicultural students.

In terms of personalization, the Social Studies AIDT did not offer adaptive learning pathways based on student comprehension or background. This omission could limit its ability to support students with varying levels of familiarity with Korean history and society (Kim & Cho, 2024). Furthermore, the feedback was general and lacked cultural relevance, while the absence of adaptive pathways hindered personalized learning experiences for students at varying levels of familiarity with Korean history and society, as shown in Figure 5. In particular, the animated video materials did not provide language options to promote understanding for multicultural students, as well the scripts were provided only in Korean language.

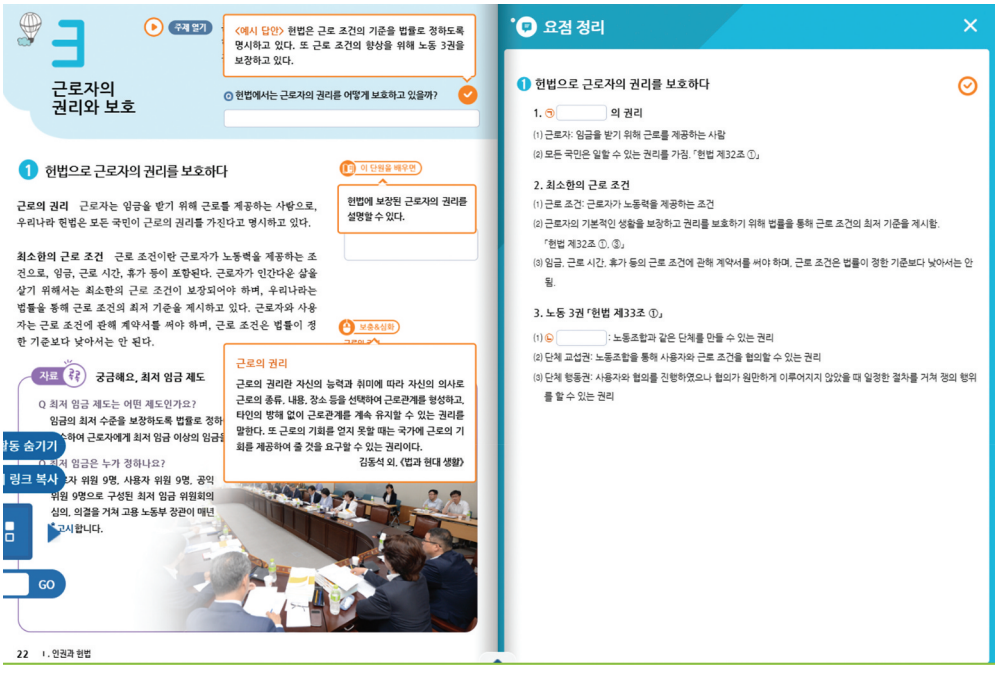


Figure 4. Key content with summary from the social studies AIDT (Kim et al., 2018, pp. 22–23).

Figure 5 highlights the use of multimedia to enhance learning. However, the analysis revealed that the feedback provided in Korean was general and lacked cultural relevance, limiting its effectiveness in supporting students from different cultural backgrounds.

Additionally, the personalized content recommendations within the Social Studies AIDT were not tailored to help students understand complex social and historical content. The absence of such tailored feedback reduces the tool's effectiveness for multicultural students.

Overall, while the Social Studies AIDT shows potential, it currently falls short in several critical areas necessary for supporting multicultural students. Improvements are needed in multilingual support, culturally relevant content, language proficiency accommodations, adaptive learning pathways, tailored feedback, and personalized content recommendations are needed to make it a more inclusive and effective educational tool.

Science for 9th graders

Science subjects emphasize hands-on learning and experimentation, frequently incorporating Sino-Korean terms and English vocabulary to explain foundational concepts. The Science AIDT incorporated animated and real experiment visuals, including augmented reality (AR) elements, to support written content. Chapter summaries and mini-tests were provided to help students review and assess their understanding, with activities requiring students to record observations and apply their knowledge, as shown in Figure 6.

The Science AIDT shares similar limitations with the English and Social Studies AIDTs in its inability to fully support multicultural students. The analysis of the Science AIDT for 9th graders revealed several critical shortcomings in supporting multicultural students. Like



Figure 5. Video support and summary from social studies AIDT (Kim et al., 2018, pp. 22–23).

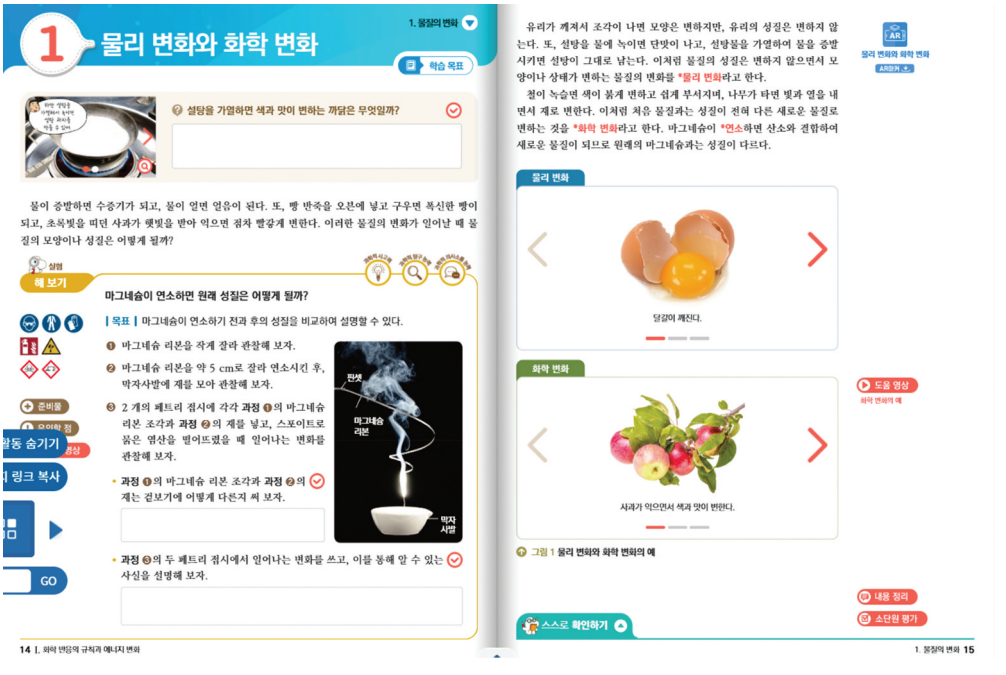


Figure 6. Key content with visual aids from the science AIDT (Kim et al., 2020 pp. 14–15).

other subjects, the Science AIDT lacked multilingual options, making it difficult for non-native speakers to understand technical and scientific terminology. Without translation tools or bilingual resources, students with limited Korean proficiency may struggle to fully comprehend the content, potentially hindering their learning outcomes.

Additionally, the Science AIDT did not provide language proficiency accommodation, such as simplified language or visual aids, to support students with limited language skills. The absence of these scaffolding techniques makes it harder for students to grasp complex scientific key concepts, leading to challenges in comprehension and engagement. This lack of support risks creating gaps in understanding, particularly for students with lower Korean proficiency.

Figure 6 illustrates the use of visual aids to support the understanding of scientific concepts. However, the analysis revealed that the textbook lacked multilingual options and adaptive content adjustments, both of which are essential for supporting students with diverse academic and language backgrounds.

Regarding personalization, the Science AIDT did not include adaptive content adjustments. This limitation means that students with varying academic and language backgrounds may struggle with the uniform material. Accessibility is crucial for addressing diverse needs, and its absence suggests the Science AIDT could not effectively support different learning paces and styles, displayed in Figure 7.

Figure 7 highlights the Science AIDT's use of self-assessment tools to help students gauge their understanding. However, the analysis revealed that the feedback provided was limited and did not account for linguistic or cultural differences. Without personalized guidance, students may lack the support needed to enhance their learning outcomes. Additionally, the science textbook did not include culturally tailored content suggestions.

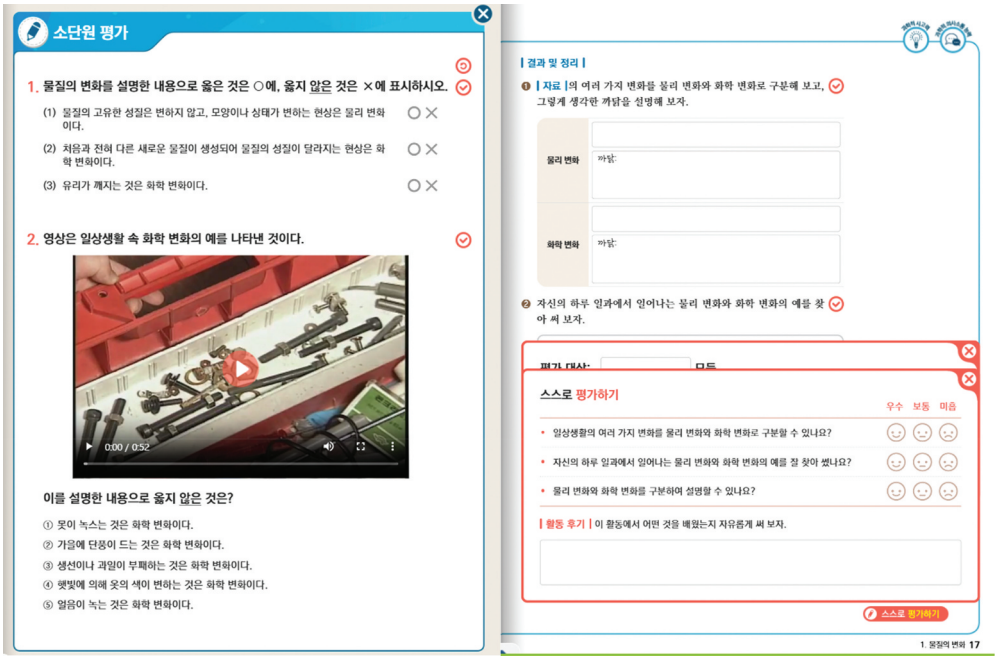


Figure 7. Chapter & self evaluation from the science AIDT (Kim et al., 2020, pp. 15 & 17).

This absence may disengage students who could benefit from seeing science through a culturally inclusive lens. Personalized content recommendations can boost engagement by connecting the material to students' own experiences and interests. The absence of this feature suggests that it may not fully support the diverse needs of its users.

Overall, while the Science AIDT shows potential, it falls short in several key areas necessary for supporting multicultural students. Improvements in multilingual support, culturally relevant content, language proficiency accommodations, adaptive learning pathways, immediate and actionable feedback, and personalized content recommendations are essential to make the textbook a more inclusive and effective educational tool.

The findings reveal that while AIDTs for English, Social Studies, and Science demonstrated promising features, they fell short in several critical areas necessary for effectively supporting multicultural students. One significant limitation across all subjects was the lack of multilingual support, as the absence of translation tools and bilingual resources created substantial barriers for non-native Korean speakers, significantly impacting accessibility. Furthermore, the failure to include culturally diverse examples diminished engagement and inclusiveness, making the material less relatable to multicultural learners. Another major shortcoming was the insufficient scaffolding, such as the lack of simplified language, visual aids, and interactive elements, which limited comprehension and created challenges for students with varying levels of proficiency. The absence of adaptive learning pathways further restricted the AIDTs' ability to meet the diverse learning paces and styles of students, as the content did not dynamically adjust based on individual performance or needs. Additionally, feedback mechanisms were generic and failed to address linguistic and cultural diversity, reducing their effectiveness in helping students track progress and improve.

These findings suggest that while AIDT holds significant promise, several key improvements are necessary to make it more inclusive and effective for multicultural learners. The lack of multilingual support, for instance, is a critical issue that hinders accessibility for non-native Korean speakers (Kim, 2020; Kim & Cho, 2024). Furthermore, the absence of culturally relevant content reduces engagement and may prevent multicultural students from feeling included in the learning process. The insufficient scaffolding and lack of adaptive learning pathways further highlight the need for AIDT to better meet the diverse needs of students (Cook & Hollebeke, 2024; Graham, 2020). Lastly, the generic feedback mechanisms emphasize the importance of creating more personalized and culturally sensitive support to help students track and improve their learning progress.

Teacher support: The Guide to Using Digital Textbooks for Middle School Teachers

The publicly accessible teacher guidebook, published in 2018 by the MoE and the KERIS, provides a comprehensive overview of AIDT content and the related teacher support materials, both within and beyond the AIDT framework. Figure 8 provides a structured overview of the teacher support materials in the AIDT, highlighting the components designed to assist educators in delivering and managing digital education.

The AIDT teacher support materials are categorized into three main sections, represented as concentric layers that illustrate their relationship within the digital textbook system: Core Content Area of Learning Materials (Green Section), Expanded Support Functions with Learning Management (Blue Section), and External Resource Integration (Purple Section).

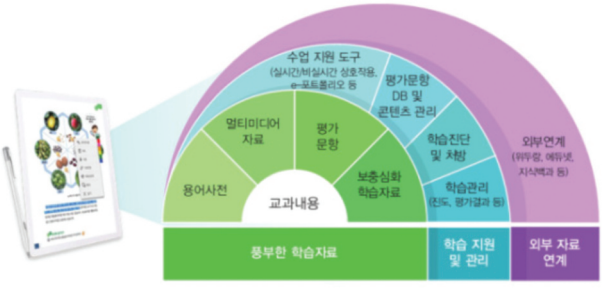


Figure 8. AIDT concept map (MoE & KERIS, 2018, p. 2).

At the centre of the *Core Content Area* is the main textbook content, providing the primary educational materials used for students. Surrounding this core are *Supplementary Learning Resources* including a glossary for key terms, multimedia materials (images, videos, interactive elements), and Assessment Questions. These resources enhance student understanding, foster engagement, and allow teachers to measure progress in real time. *Cultural and Social Contextual Materials* are also integrated to support inclusive education by relating to students' diverse backgrounds.

The *Expanded Support Functions* section includes tools for classroom management and student assessment. These tools enable teachers to evaluate student performance, manage digital content, diagnose learning gaps, and track progress. The system also simplifies workflow by organizing assignments and educational activities, helping teachers tailor instruction to individual needs.

The outermost section, *External Resource Integration*, connects the digital textbook to additional educational content, such as community resources, external services, and evaluation tools. These features broaden the learning experience, providing students with access to local cultural materials and supplemental education resources, which enrich learning beyond the core content.

In summary, the teacher's guide provides a well-rounded system with core content, supplementary resources, and tools for student assessment and classroom management. The integration of multimedia and culturally contextual materials allows for flexible lesson creation, while expanded support functions assist teachers in tracking student progress and tailoring instruction. The connection to external resources enhances learning by offering additional cultural and social context.

Table 1 summarizes the analysis, showing that while AIDT has potential, the current test versions lack critical features for supporting multicultural students, such as multilingual support, cultural relevance, adaptive pathways, and tailored feedback. Enhancing these areas is essential for AIDT to fulfil its promise of inclusive, personalized education, especially as Korea's educational demographics become increasingly diverse.

The literature highlights AIDT's potential to address challenges faced by multicultural students, particularly those related to language barriers and differentiated instruction (Choi, 2024; Kim, 2020; Kim & Cho, 2024). However, the current design often overlooks the needs of these learners. For example, while personalized feedback and interactive content benefit native Korean students, they do not account for the varying language proficiency levels among multicultural learners. To better serve all students, integrating multilingual support

Table 1. Summary of AIDT analysis based on criteria.

	Content Accessibility			Diversity & Personalization			Support	
	Multilingual Support	Culturally Relevant Content	Language Proficiency Accommodation	Adaptive Learning Pathways	Interactive Feedback	Personalized Content Recommendations	AI Tutor Functionality	Assistant Teacher Tools
English	– No translation tools – No bilingual resources	– Minimal multicultural examples/ activities – Content lacks diverse perspectives	– Minimal scaffolding techniques – Few visual aids or simplified language	– No dynamic adjustment for difficulty – Content not tailored to language proficiency	– Limited, mainly textual – No cultural context provided	– Lacks culturally relevant suggestions – Content not tailored to students' backgrounds	– No interactive, tailored support	– Few resources for multicultural classrooms – Limited data-driven lesson aids
Social Studies	– No multilingual support – Limited accessibility for non-Korean speakers	– No diverse cultural representation – Limited inclusivity for multicultural students	– No language support features – No simplified language or aids for complex concepts	– Not adapted for different academic backgrounds – Single, uniform difficulty level	– Minimal, general feedback – Feedback lacks specific cultural relevance	– Lacks content aligned with diverse backgrounds – Limited relevance for multicultural students	– No tailored explanations – Limited AI-driven academic support	– Few resources for cultural inclusivity – Lacks adaptive aids for diverse classroom needs
Science	– No multilingual options – Technical terms only in Korean	– No reference to diverse contributions to science – Limited relevance for students from varied backgrounds	– No simplified language for complex terms – Lacks visual aids	– No adaptive pathways – Uniform content complexity	– Minimal, non-specific feedback – Limited to standard academic context	– No tailored content for diverse student needs – Science examples do not reflect cultural inclusivity	– No interactive AI support for learning – Limited engagement for multicultural students	– Lacks resources for handling diverse student needs – Minimal teacher aid for science concepts adapted to multicultural contexts

and culturally responsive content is crucial. Furthermore, comprehensive teacher training is essential to help educators use these tools and support multicultural learners.

AIDT can offer students diverse educational opportunities, moving beyond traditional memorization and overcoming limitations of hands-on experiments. However, if developed solely for Korean students, AIDT could exacerbate educational inequality. To ensure equitable access for all students, it is vital to address the specific needs of culturally and linguistically diverse (CLD) students, focusing on access, content relevance, and instructional support.

While the AIDT teacher support materials offer comprehensive tools for fostering an inclusive and adaptive learning environment, several limitations remain. First, the core content's limited multilingual support and lack of cultural depth reduce its effectiveness for linguistically and culturally diverse students. The teacher's guide does not provide readily accessible additional language materials, which limits its applicability in diverse classrooms. Second, the reliance on external resources to introduce cultural relevance may overwhelm teachers who lack sufficient training. Many of the resources in the guide require teachers to process and adapt the materials according to their specific needs, which can be time-consuming. Additionally, the system's adaptive learning capabilities focus more on diagnostics than on content customization, requiring teachers to learn how to adjust the system themselves, which could be time-intensive. Lastly, AIDT's dependence on digital access may exacerbate issues of equity, particularly in low-resource settings, where students may lack the necessary technological infrastructure.

Conclusion

In conclusion, while AIDT shows significant promise in supporting multicultural students, further research and pilot programmes are essential to assess its long-term impact in diverse classrooms. It is essential to monitor how these tools perform in real-world settings, particularly in schools with high multicultural populations, to ensure that the technology meets the needs of all students. Collaborative efforts between policymakers, educators, and developers are necessary to refine AIDT and make it an inclusive tool that fosters academic success for multicultural students in Korea and beyond.

Additionally, teacher training is essential to effectively integrate these elements. Educators need professional development focused on technology use, cultural competencies, and differentiated instruction to foster inclusive environments. Without these enhancements, disparities between Korean and multicultural students will widen, leaving immigrant-background students behind, both in Korea and their home countries. Brown's study (2024) on linguistic and cultural homogeneity in Korea emphasizes the challenges multicultural students face, reinforcing the need to adapt educational tools to support diverse needs.

Recommendations

Based on the findings and analysis presented in this study, several strategies are necessary to enhance the effectiveness of AIDT and ensure their successful use in multicultural middle school classrooms. While AIDT holds significant potential for supporting diverse student populations, there are key areas where it must be improved to fully address the unique needs of multicultural learners. Additionally, teacher training plays a critical role in the successful integration of these technologies,

as educators must be equipped with tools and knowledge to create inclusive learning environments. Therefore, we present the following recommendations in two main areas: 1) enhancing AIDT features, and 2) strengthening teacher training, both in the short-term and long-term, to ensure effective Information and Communication Technology (ICT) integration in multicultural classrooms.

Enhancing AIDT features for multicultural students

Based on the analysis of AIDT's current capabilities and limitations, several features need to be enhanced to ensure that the tool effectively supports multicultural students. While AIDT has the potential to offer personalized, interactive learning experiences, its current design does not fully address the diverse needs of multicultural learners. Key improvements are necessary in areas such as language support, cultural relevance, and adaptability to individual learning needs. By incorporating these enhancements, AIDT can become a more inclusive and effective tool for students from diverse linguistic and cultural backgrounds. The following recommendations aim to refine AIDT's features, making them more responsive to the unique challenges faced by multicultural middle school students.

(1) **Real-time Feedback**: Essential for identifying and correcting misunderstandings quickly, especially in high-paced learning environments. AI-driven feedback can offer language tips and alternative strategies, supporting both Korean and multicultural students in subjects like math and grammar. Over the long term, real-time feedback can play a critical role in fostering academic success for both Korean and multicultural students.
(2) **Interactive Simulations**: These provide immersive learning experiences, helping students grasp complex concepts through virtual hands-on practice. Simulations can initially focus on language courses and gradually expand into Science, Technology, Engineering, and Mathematics (STEM) subjects, offering language support and practical experiences.
(3) **Personalized Study Plans**: AI can create customized plans that adjust based on students' progress. For Korean students, this ensures focus on areas needing improvement, while for multicultural students, it accommodates evolving language proficiency. Personalized plans reduce repetition, boosting efficiency and student success.
(4) **Collaborative Learning Tools**: These foster teamwork and social skills are essential in diverse classrooms. For multicultural students, collaboration tools help them develop interpersonal skills in a new environment. In the short-term, teacher-led collaboration can transition to more student-driven projects over time.
(5) **Enhanced Cultural Understanding through Group Projects**: Projects that encourage students to share insights from their cultural backgrounds can enhance inclusivity and cultural understanding. Short-term, these can be teacher-led; long-term, they can be facilitated through AI with tailored support for teachers.

Incorporating these enhancements into AIDT will significantly improve its ability to support multicultural students, providing them with the tools they need to succeed in an increasingly diverse classroom environment. By focusing on real-time feedback, interactive simulations, personalized study plans, collaborative learning tools, and enhanced cultural

understanding, AIDT can become more adaptive and responsive to the needs of students from varied linguistic and cultural backgrounds. These improvements are crucial for ensuring that all students, regardless of their background, have equitable access to high-quality education. In the next section, we will explore the critical role of teacher training in effectively integrating these features into the classroom and ensuring their sustainable use.

Teacher training for effective ICT integration in multicultural classrooms

Teacher training is essential to the successful implementation of AIDT. To maximize the benefits of AI and digital tools, educators need structured, tailored training that considers the unique dynamics of multicultural classrooms (UNESCO, 2018). **Short-term training** should focus on building foundational ICT skills, such as using real-time feedback tools, collaborative platforms, and multilingual resources to support diverse learners. In addition, this training must address strategies for fostering inclusive environments where all students feel represented. **Long-term professional development** programmes can build on these basics by incorporating innovative teaching strategies, such as designing culturally responsive lessons, creating AI-driven lesson plans, and leveraging adaptive technologies to meet the varied needs of multicultural learners (Cook & Hollebeke, 2024).

For teachers to effectively integrate ICT into their classrooms, they must not only master the technical aspects of these tools but also understand how they can be used to enhance learning for students from diverse linguistic and cultural backgrounds (eCampusOntario, n.d.). This dual focus on both technical proficiency and pedagogical awareness will ensure that teachers are equipped to use ICT in ways that promote equity, inclusion, and academic success for all students.

Short-term training goals

In the short term, teacher training programmes should prioritize building fundamental ICT skills. These foundational skills include proficiency in:

- **Real-time feedback tools**: Teachers can use AI-driven platforms that provide instant feedback to students, helping them track progress and address learning gaps efficiently.
- **Collaborative platforms**: Digital tools that facilitate collaboration, such as shared online workspaces, can promote teamwork and peer learning across diverse student groups.
- **Multilingual resources**: Familiarity with language-supportive technologies, such as translation tools and multilingual interfaces, is crucial for addressing language barriers and ensuring equitable access to learning materials.

Additionally, short-term training should focus on equipping teachers with strategies for creating inclusive environments. This can be achieved by promoting cultural sensitivity and awareness in lesson planning and classroom interactions, using digital storytelling tools to incorporate diverse cultural narratives, and encouraging student participation through interactive and culturally responsive activities. These short-term initiatives aim to immediately enhance teachers' ability to integrate ICT tools effectively while fostering a sense of inclusion and belonging among students.

Long-term professional development

Over time, professional development programmes should expand to encompass more advanced and innovative approaches to teaching in multicultural classrooms. Key areas of focus include:

- **Innovative teaching strategies**: Teachers should learn to design lesson plans that leverage AI to personalize instruction, adapt to different learning styles, and meet the specific needs of multicultural learners.
- **Culturally responsive pedagogy**: Long-term training should guide educators in embedding cultural relevance into AI-driven tools, ensuring that teaching materials reflect the diverse experiences and backgrounds of students.
- **Adaptive technologies**: Teachers must be trained to use adaptive learning systems that analyse student performance data and provide tailored recommendations to address individual challenges and strengths.

Furthermore, sustained professional development programmes should emphasize continuous learning and collaboration among educators. This can be achieved through peer mentoring and sharing best practices within multicultural contexts, participation in workshops and conferences focused on ICT and multicultural education, and engagement in research and pilot programmes to explore emerging technologies in education.

By addressing both short-term and long-term training needs, teachers will be better prepared to lead classrooms enriched by ICT. Structured training enables educators to create learning environments that not only incorporate advanced technologies but also prioritize equity and inclusion (UNESCO, 2018). This dual focus ensures that students from all cultural and linguistic backgrounds have access to high-quality education, bridging gaps and fostering mutual understanding. As the global education landscape continues to evolve, the role of teachers as facilitators of technology and inclusion will become increasingly necessary. Investing in their training is an essential step towards building classrooms where every student has the opportunity to succeed.

Disclosure statement

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